Syllabus: Vapour absorption (VAR) system – description and working of NH_3-water, LiBr-water (two shell and four shell), maximum COP, three-fluid absorption system. Steam jet refrigeration – principle, components, estimation of motive steam. Thermo-electric refrigerator, vortex (Hilsch) tube.

I. Vapour Absorption Refrigeration (VAR) System

- Heat energy (steam, hot water, waste heat, solar, gas flame) drives the system instead of mechanical work. The compressor of VCR is replaced by the **absorber + pump + generator**.
- Two working substances: **refrigerant** (evaporates) and **absorbent** (absorbs the refrigerant vapour). Pair examples: NH_3 (refrigerant) – water (absorbent); water (refrigerant) – LiBr (absorbent).

Principle

- Refrigerant vapour at low pressure is **absorbed** in the weak solution (exothermic, cooled by water). Absorbing needs far less energy than compressing – a liquid can be pumped with 1-2 % of the work needed for vapour.
- Strong solution is pumped to high pressure and **heated** in the generator to boil off refrigerant vapour. This vapour goes to the condenser, throttle valve, evaporator – the same as in VCR.

> Compressor in VCR ≈ Absorber + Pump + Generator (+ heat exchanger) in VAR

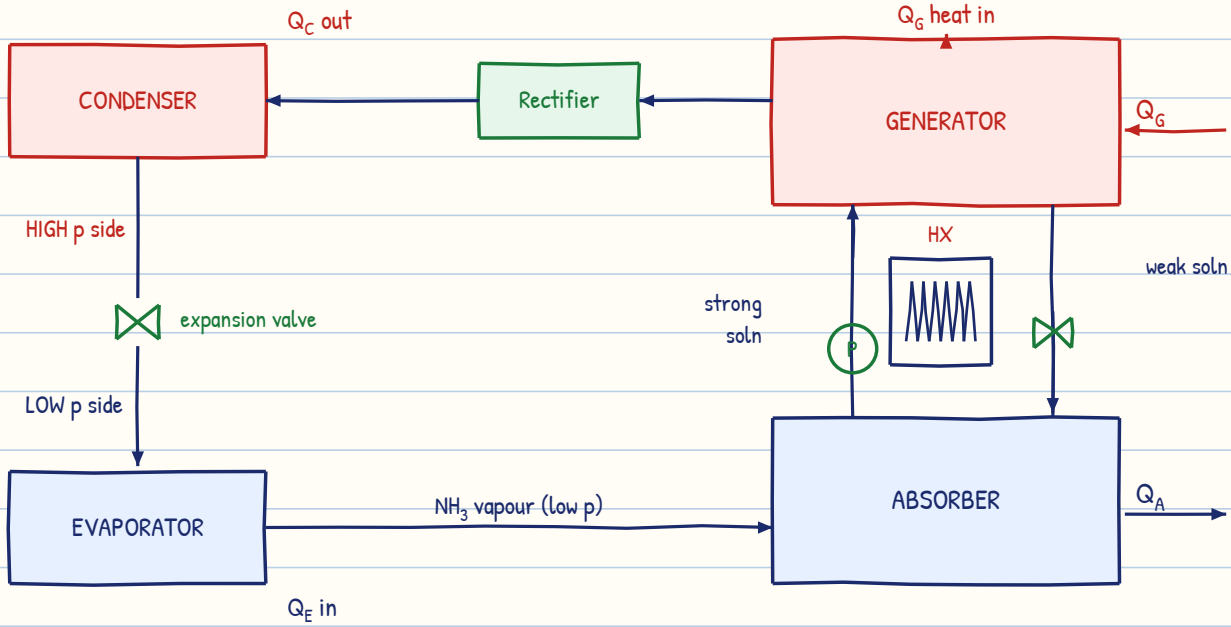
Advantages / disadvantages over VCR

- **Pros:** uses low-grade heat / waste heat; no moving parts except small pump → quiet, less vibration, reliable; capacity control easy (from 100 % to 10 %); no ODP / GWP refrigerant.
- **Cons:** low COP (0.5 - 1.2) so larger heat rejection; bulky and costly; NH_3 is toxic; LiBr crystallises; needs cooling tower.

Property	VCR	VAR
Energy input	Work (electricity)	Heat (steam, gas, waste)
Moving parts	Compressor (many)	Only small pump
COP	3 - 5	0.5 - 1.2

Property	VCR	VAR
Noise / vibration	Higher	Low
Part-load	Poor	Good
Size & cost	Smaller	Bigger, costlier
Best use	Everywhere	Where cheap heat is available

2. NH_3 – Water Absorption System (Description and working)



Components and functions

- **Evaporator:** liquid NH_3 at low p evaporates, absorbing Q_E from the cold space.
- **Absorber:** NH_3 vapour is absorbed by cold weak solution (water poor in NH_3) → strong solution. Heat of absorption Q_A is removed by cooling water. Low T of absorber helps absorption.
- **Pump:** raises pressure of strong solution from p_e to p_c (small work).
- **Heat exchanger:** strong solution (cold) picks up heat from the weak solution (hot) → saves generator heat and cools the weak solution before absorber.
- **Generator:** strong solution heated (steam / gas) and NH_3 vapour driven out. Water also vaporises a little as the boiling points are close.
- **Analyser and Rectifier (dephlegmator):** remove water vapour from NH_3 vapour (water vapour would freeze in the evaporator and choke the valve). Water condenses and drains

back to generator. NH_3 reaching condenser is about 99.8 % pure.

- **Condenser:** NH_3 vapour condensed by cooling water. **Expansion valve:** throttles the liquid to p_e.

Working in sequence

- Pure NH_3 vapour from generator → rectifier → condenser (Q_C rejected) → liquid → expansion valve → evaporator (Q_E absorbed) → vapour → absorber → strong solution → pump → heat exchanger → generator. The weak solution flows back through the heat exchanger and pressure-reducing valve to absorber.

> **Note:** In NH_3-water, the refrigerant (NH_3) and the absorbent (water) are both volatile → rectifier is necessary. In LiBr-water, LiBr is non-volatile → no rectifier is needed.

Practical improvements

- Heat exchanger between strong and weak solution; sub-cooler (liquid-vapour HX) to raise RE; rectifier with analyser to purify; proper cooling of absorber.

3. Maximum COP of an Absorption System

Model the system as a reversible **heat engine** (between T_G and T_0) driving a reversible **refrigerator** (between T_E and T_0). Three temperatures: generator T_G, surroundings (condenser & absorber) T_0, evaporator T_E.

$$COP_{actual} = Q_E / Q_G \text{ (pump work neglected)}$$

$$\text{Engine efficiency } \eta_{max} = (T_G - T_0) / T_G \text{ ; Refrigerator COP} = T_E / (T_0 - T_E)$$

$$\mathbf{COP_{max} = (T_G - T_0) / T_G \times T_E / (T_0 - T_E)}$$

- Maximum COP rises with higher T_G, higher T_E, lower T_0. Real COP is 60-80 % of this.

Example 1

T_G = 100 °C (373 K), T_0 = 30 °C (303 K), T_E = −10 °C (263 K):

$$COP_{max} = (70/373) \times (263/40) = 0.1877 \times 6.575 = \mathbf{1.23}$$

Example 2

A NH_3-water plant (T_G = 90 °C, T_0 = 35 °C, T_E = −5 °C): COP_{max} = (55/363) × (268/40) = 0.1515 × 6.7 = **1.015**. Heat needed for 10 TR at actual COP 0.6: Q_G = 35.17/0.6 = 58.6 kW.

4. LiBr – Water Absorption System

- Refrigerant = **water**, absorbent = **lithium bromide** (a salt, non-volatile). Evaporator works at ≈ 6 mm Hg abs (≈ 4 °C), so the plant is under high vacuum. Used for **air-conditioning and chilled water** (T_e > 0 °C only, as water freezes).
- Absorbent properties: strong affinity for water vapour, stable, non-toxic, no rectifier needed. Problems: **crystallisation** at high concentration (> 65 %) and low temperature; corrosive – needs inhibitors; vacuum maintained by purge unit.

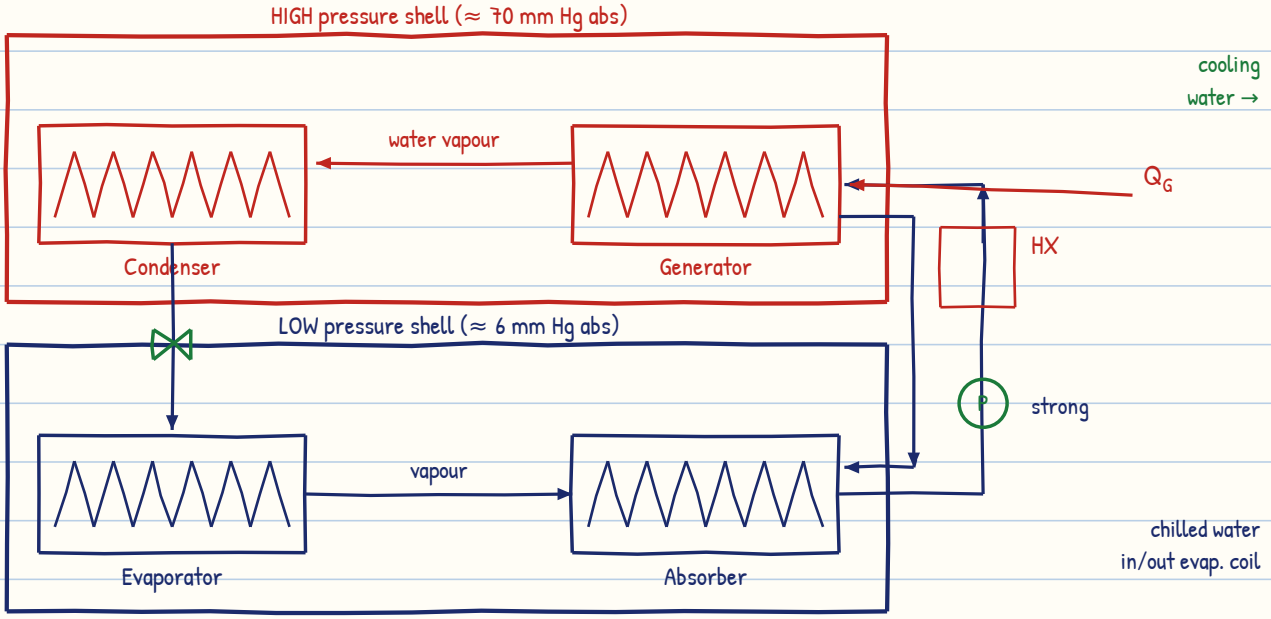
Solution concentration and Dühring chart

- The state of LiBr-water solution is plotted on a **Dühring chart** (solution temperature vs refrigerant saturation temperature, lines of constant concentration) – it shows how the same vapour pressure needs a higher temperature as the concentration rises.
- Typical cycle: absorber outlet ≈ 58 % LiBr at 35-40 °C (dilute); generator outlet ≈ 63-64 % at 90-100 °C (strong). The difference in concentration is called the **degassing range**; limits crystallisation.
- **Crystallisation control:** keep the concentration below the crystallisation line; dilution cycle at shutdown; avoid sudden drop in cooling water temperature.

Double-effect LiBr-water

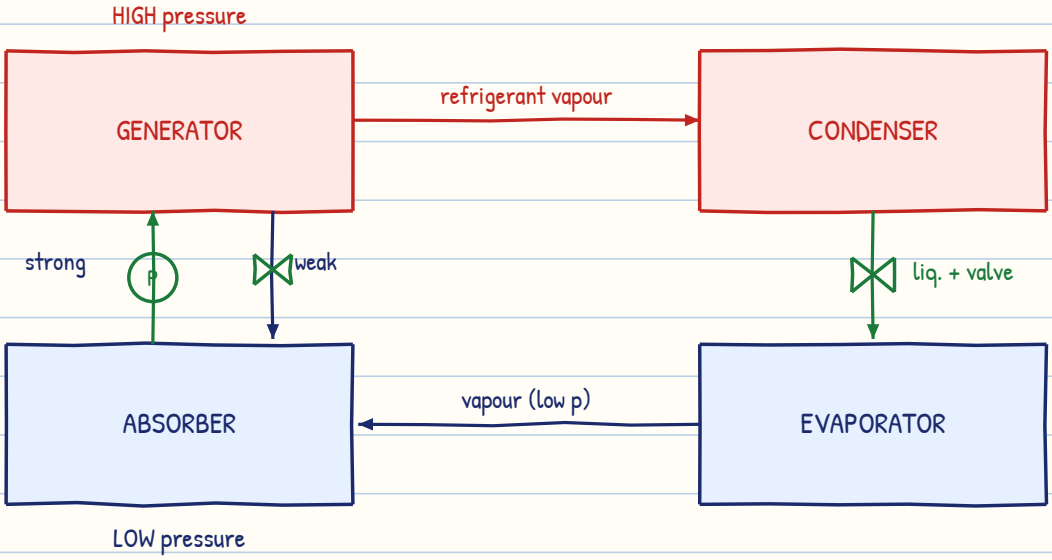
- Two generators: the high-temperature generator (≈ 150 °C, driven by steam at 8 bar or gas flame) boils vapour that is then used as the **heating medium of the low-temperature generator**; the heat is used twice → COP ≈ 1.0-1.3.

Two-shell system



- ***Upper (high-pressure) shell:*** *generator + condenser.* ***Lower (low-pressure) shell:*** *evaporator + absorber.*
- *Dilute solution from absorber is pumped through the heat exchanger to the generator; steam / hot water boils water vapour out of it. Vapour condenses (cooling water), liquid water is throttled (U-tube / orifice) to evaporator and sprayed on chilled-water coil – evaporates and cools the chilled water (e.g. 12 → 7 °C).*
- *Concentrated solution returns through HX to the absorber and is sprayed over the cooling coil; it absorbs the water vapour and is diluted. Cooling water goes first to absorber, then condenser.*
- *Most common type for small-medium sizes (single-effect, COP ≈ 0.7).*

Four-shell system



- Generator, condenser, evaporator and absorber are placed in **four separate shells** connected by pipes. It allows flexible layout, easy maintenance and use for very large capacity or **double-effect** chillers (two generators, COP ≈ 1.2).
- Used where height is limited or when higher temperature heat source is available (direct-fired). Costlier due to more shells and piping.

Feature	Two shell	Four shell
Shells	2 (HP & LP)	4 (each component)
Size & cost	Compact, cheaper	Large, costlier
Maintenance	Harder (combined)	Easy
Use	Small & medium	Large / double-effect

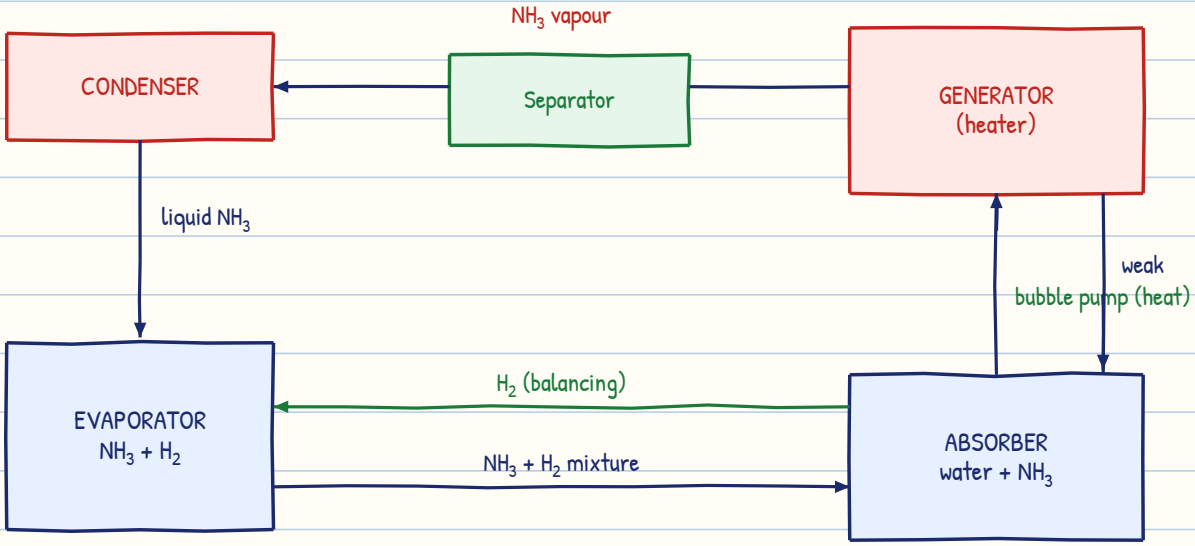
NH_3-water vs LiBr-water

NH_3-water	LiBr-water
Refrigerant volatile → rectifier	Absorbent non-volatile → no rectifier
Can reach below 0 °C	Only above 0 °C (water refrigerant)
Works at above atm pressure	Works under high vacuum
Toxic (NH_3)	Non-toxic
Cools industrial, cold storage	Used for air conditioning
COP ≈ 0.5	COP ≈ 0.7 - 1.2

> **Note:** Water has a very high latent heat (2500 kJ/kg), hence LiBr-water has better COP than NH_3-water.

5. Three-Fluid Absorption System (Platen-Munters)

- Invented by Platen and Munters (Sweden). Three fluids: **NH_3** (refrigerant), **water** (absorbent) and **hydrogen** (inert gas that maintains the **total pressure equal** throughout the plant).

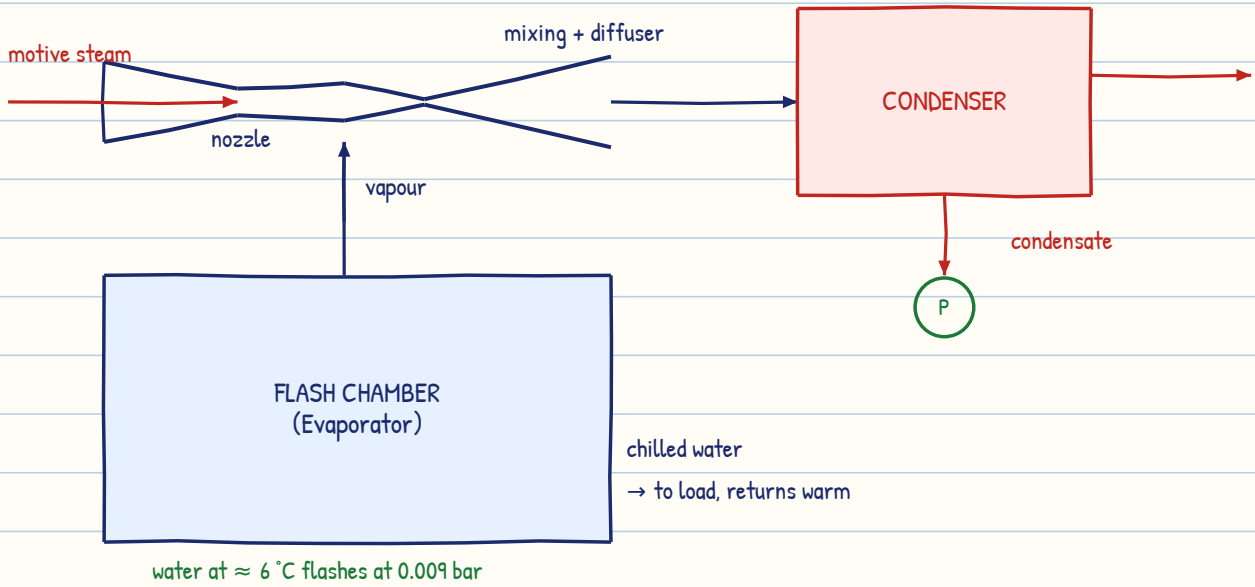


Total pressure same everywhere – no pump, no valve

- Total pressure is the same everywhere (≈ 15 bar), so **no pump and no expansion valve** are needed – circulation is by density difference (**bubble pump / thermosyphon**) and gravity. NH_3 evaporates in the evaporator because its **partial pressure** is lowered by H_2: $p_{total} = p_{NH3} + p_{H2}$.
- Heat from electric heater / kerosene flame lifts weak solution by bubble pump; NH_3 vapour condenses; liquid NH_3 flows to evaporator where it diffuses into H_2 and absorbs heat; the NH_3-H_2 mixture goes to absorber where water absorbs NH_3, and H_2 returns.
- **Uses:** silent domestic refrigerators, hotel mini-bars, caravans (LPG) – no moving parts. COP is low (≈ 0.2-0.3).

6. Steam Jet Refrigeration System

- Water is used as the refrigerant! Water is evaporated at a very low pressure (≈ 0.009 bar at 6 °C) in a **flash chamber**; latent heat comes from the remaining water, which is **chilled**. High pressure steam acts through an **ejector** to maintain the vacuum and to compress the vapour.



Basic components

- **Flash chamber (evaporator):** warm return water flashes at low p; unflashed water is chilled to 4-10 °C.
- **Ejector:** convergent-divergent **nozzle** (steam expands to p_e, high velocity ≈ 1000 m/s), **suction / mixing chamber** (entrains vapour from flash chamber) and **diffuser** (velocity → pressure).
- **Condenser:** condenses motive steam + flash vapour (surface or barometric type). **Air ejector** removes air; **hot well & pumps** return condensate to boiler and make-up water to flash chamber.

Working

- Steam at 8-10 bar → nozzle (p_e ≈ 0.01 bar) → entrains flash vapour → diffuser raises p to condenser (≈ 0.05 bar) → condensed. Chilled water is circulated to the load (air handler).

Estimation of motive steam

- Fraction flashed: $x = c_{pw} (T_{in} - T_{out}) / h_{fg}$. Flash vapour $m_v = Q_L / h_{fg}$ (when cooling duty Q_L is given).

- Energy balance on nozzle / diffuser: kinetic energy of steam jet is shared with the entrained vapour:

$$m_s\,(h_s - h_n)\,\eta_n\,\eta_d = m_v\,(h_d - h_v) \;[\text{approx.}]$$

Entrainment ratio $ER = m_v/m_s$; motive steam $m_s = m_v/ER$ ($ER \approx 0.3 - 0.6$)

Example

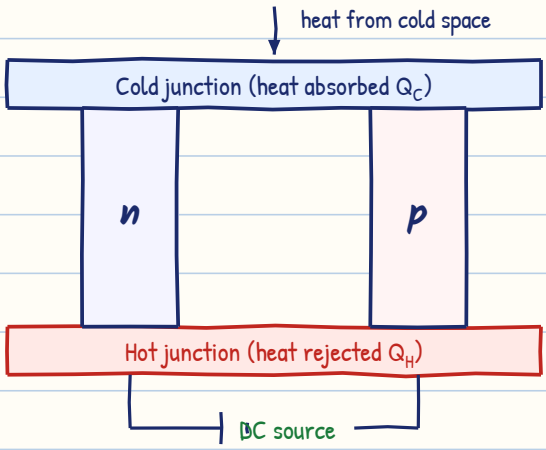
Cooling load 10 TR at 6 °C (h_{fg} = 2487 kJ/kg): $m_v = 35.17/2487 = 0.01414$ kg/s. If ER = 0.5, $m_s = 0.0283$ kg/s = **102 kg/h** ≈ 10 kg/h per TR.

Merits and demerits

- Merits: no moving parts, no harmful refrigerant, cheap, uses waste steam; handles large loads.
- Demerits: low COP (≈ 0.3), high steam consumption, can't go below 0 °C, ejector designed for one load, need cooling water.

7. Thermo-electric Refrigerator

- Based on the **Peltier effect** (reverse of Seebeck): when current passes through a circuit of two dissimilar conductors (n-type and p-type semiconductor), one junction cools and the other heats.



Peltier effect: when DC passes through a junction of two dissimilar semiconductors (Bi_2Te_3), one junction cools, other heats.

$Q_C = I\,T_C - ½\,I^2R - K\,\Delta T$

$W = I\,\Delta T + I^2R$

- Electrons in n-type material carry heat from cold junction to hot junction; semiconductors (Bi_2Te_3) have high Seebeck coefficient , low thermal conductivity K and low resistance R (figure of merit $Z = {}^2/(K\,R)$).

- Heat sink (fins / fan) on hot side. Many couples in series (thermopile) give higher capacity.

$$Q_C = \alpha I T_C - \tfrac{1}{2} I^2 R - K (T_H - T_C) \; ; \; W = \alpha I (T_H - T_C) + I^2 R \; ; \; COP = Q_C / W$$

Example

Module with $\alpha = 0.05$ V/K, $R = 2\ \Omega$, $K = 0.5$ W/K, $T_C = 273$ K, $T_H = 303$ K, $I = 2$ A:

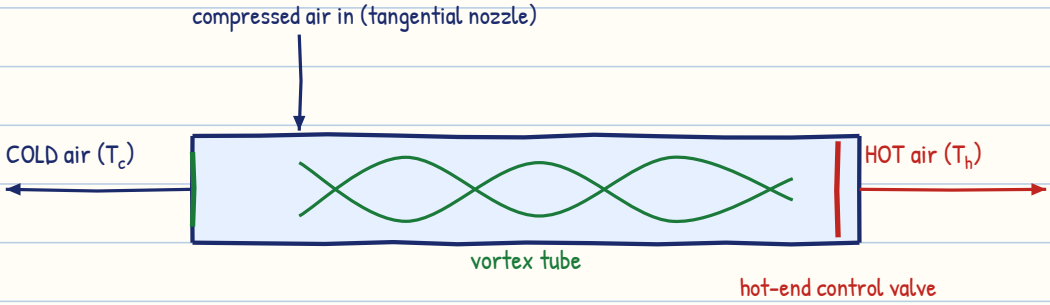
$$Q_C = 0.05 \times 2 \times 273 - 0.5 \times 4 \times 2 - 0.5 \times 30 = 27.3 - 4 - 15 = \mathbf{8.3\ W}$$

$$W = 0.05 \times 2 \times 30 + 4 \times 2 = 3 + 8 = 11\ W \; ; \; COP = 8.3/11 = \mathbf{0.75}$$

- **Advantages:** no moving parts, silent, compact, no refrigerant, reversible (heating also), precise control. **Disadvantages:** low COP, high cost, needs DC, small capacity. **Uses:** portable coolers, car mini-fridge, electronic chip cooling, medical, space, water coolers.

8. Vortex Tube (Hilsch Tube)

- Invented by Ranque, developed by Hilsch. No moving parts: compressed air (5–7 bar) is blown **tangentially** into a tube, producing a rotating vortex. The outer layers become **hot** and leave through the hot-end valve; the inner core becomes **cold** and leaves through the orifice at the other end.



$$\mu_c T_c + (1 - \mu_c) T_h = T_i$$

μ_c = cold mass fraction = m_c / m_i

- **Why separation occurs:** (i) the free vortex rotates faster at the centre, so the core loses energy by shear to outer layers (angular momentum, friction); (ii) expansion cools the core, work done on the outer layers heats them.

- Cold mass fraction $_c = m_c/m_i$. Cold-end orifice gives lowest T when $_c \approx 0.3$; maximum cooling effect ($_c \cdot \Delta T_c$) at $_c \approx 0.6$-0.7.

> Energy balance: $_c T_c + (1 - _c) T_h = T_i$

Example

Inlet air 300 K, $_c = 0.4$, $T_c = 270$ K → $T_h = (300 - 0.4\times270) / 0.6 =$ **320 K**. Refrigeration effect = $m_c c_p (T_i - T_c) = 0.4\ m \times 1.005 \times 30 = 12\ m$ kJ/s.

- **Advantages:** simple, cheap, no moving parts, maintenance free, instant cooling, compact. **Disadvantages:** low efficiency (COP ≈ 0.1), noisy, needs compressed air. **Uses:** spot cooling of machine tools, soldering, electronic cabinets, CNC machining, cooling of suits.

Problem 1 (heat balance of absorption plant)

A VAR plant gives 50 kW cooling with actual COP 0.7. Find generator heat and heat rejected.

> $Q_G = 50/0.7 =$ **71.4 kW** ; heat rejected (condenser + absorber) = $Q_G + Q_E = 71.4 + 50 =$ **121.4 kW** (pump work neglected)

> Cooling water needed for $\Delta T = 6$ K: $m = 121.4/(4.187\times6) =$ **4.83 kg/s**

Problem 2 (steam jet)

10 kg/s of water is chilled from 12 °C to 6 °C in a flash chamber ($h_{fg} = 2487$ kJ/kg). Entrainment ratio 0.4.

> Capacity = 10 × 4.187 × 6 = 251.2 kW = **71.4 TR** ; flash vapour = 251.2/2487 = 0.101 kg/s

> Motive steam = 0.101/0.4 = 0.2525 kg/s = **909 kg/h** → 12.7 kg/h per TR

Problem 3 (vortex tube)

Air at 400 kPa, 300 K enters; $_c = 0.3$, cold air leaves at 255 K. Find hot air temperature:

> $T_h = (300 - 0.3\times255)/0.7 = (300 - 76.5)/0.7 =$ **319.3 K**

Comparison of special refrigeration systems

System	Energy input	COP	Special feature
Three-fluid absorption	Heat (flame / heater)	0.2 - 0.3	No moving parts, H_2 balances pressure
Steam jet	High-pressure steam	0.2 - 0.4	Water refrigerant, ejector
Thermo-electric	DC electricity	0.3 - 0.8	Solid state, Peltier effect
Vortex tube	Compressed air	0.1 - 0.2	No moving parts, spot cooling

9. Quick Revision and Exam Questions

- $COP_{max} = (T_G - T_0)/T_G \cdot T_E/(T_0 - T_E)$; VAR uses heat; absorber + pump + generator replace the compressor.
- NH_3-water needs rectifier; LiBr-water does not (non-volatile) but needs vacuum, T > 0 °C.
- Three-fluid: NH_3 + water + H_2, no pump, same total pressure.
- Steam jet: nozzle – mixing – diffuser; water refrigerant; ≈ 10 kg steam/h per TR.
- Peltier: $Q_C = IT_C - ½I^2R - K\Delta T$. Vortex: $_cT_c + (1-_c)T_h = T_i$.

Short answers (2 marks)

- **What is the function of a rectifier?** Removes water vapour from NH_3 vapour so that only pure NH_3 enters the condenser.
- **Why no rectifier in LiBr system?** LiBr is non-volatile; only water vapour leaves the generator.
- **What is entrainment ratio?** Mass of vapour sucked per kg of motive steam.
- **Why is hydrogen used in Platen-Munters?** To equalise total pressure and lower the partial pressure of NH_3 so it can evaporate without a pump or valve.
- **What is Peltier effect?** Heating or cooling at the junction of two dissimilar conductors when current passes through.
- **Cold mass fraction?** Ratio of cold air mass flow to the total air flow in a vortex tube.
- **Why does LiBr plant need vacuum?** Water has to evaporate at about 4 °C, which needs 6 mm Hg absolute.

Likely questions

- Describe the NH_3-water absorption system with a neat sketch; role of rectifier, analyser. (long)
- Explain LiBr-water two-shell / four-shell systems; compare with NH_3-water. (long)
- Derive the expression for maximum COP of an absorption system. (long)
- Explain the three-fluid absorption refrigeration system. (long)
- Steam jet refrigeration: working, components, estimation of motive steam. (long)
- Thermo-electric refrigerator and vortex tube – working, advantages. (short / long)